BIFX 546: Machine Learning for Bioinformatics

Project Progress Check-In

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Dataset: Parkinsons Telemonitoring

Project Overview:

This document reports the preliminary exploratory data analysis (EDA) for the **Parkinsons Telemonitoring** dataset from the UCI Machine Learning Repository. The dataset contains **5,875 voice recordings** from **42 patients** with early-stage Parkinson's disease, tracked over approximately six months.

Each row corresponds to one recording and includes:

- Demographics: **Subject#**, **Age**, **Sex**, **Test_Time** (days since recruitment)
- Clinical targets: **Motor_UPDRS**, **Total_UPDRS**
- 16 biomedical measures **(Jitter**, **Shimmer**, **NHR**, **HNR**, **RPDE**, **DFA**, **PPE**, etc.).

Main long-term goal is to predict **Motor_UPDRS** and **Total_UPDRS** scores from the voice features as a regression problem and understand which voice measures track disease severity over time.

Exploratory Data Analysis:

A. <u>Data Loading and Initial Cleaning</u>

```
# ============================================================================
# A. DATASET LOADING AND CLEANING
# ============================================================================

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```python
# Load the dataset
df = pd.read_csv('Pakisons-Data-File.csv')

# Basic structure
df_shape = df.shape
n_patients = df["subject#"].nunique()
n_records = len(df)

# Check for missing values
missing_values = df.isnull().sum()

# Check for duplicates
duplicates = df.duplicated().sum()

# Check data types
print(df.dtypes)

# Drop the unnecessary index column
if "Unnamed: 0" in df.columns:
    df = df.drop(columns=["Unnamed: 0"])
df.columns
```
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The Parkinsons Telemonitoring dataset has been successfully loaded into Python using pandas. The dataset contains:
- 5,875 voice recordings from 42 patients with early-stage Parkinson's disease
- 22 features including demographics, UPDRS scores, and 16 voice measurements
- Recordings span approximately 6 months per patient (~140 recordings per patient on average)

Initial Data Cleaning
Data quality checks revealed:
- No missing values - All 5,875 rows are complete across all 22 columns
- No duplicate recordings - Each row represents a unique voice recording
- Appropriate data types - All features correctly encoded (4 integers, 19 floats)

The file includes an extra index column "Unnamed: 0" created during export, which was drop before analysis.

B. Summary Statistics

```python
# ==========================================================================
# B. SUMMARY STATISTICS
# ==========================================================================
# Basic descriptive statistics
summary = df.describe()
```

```
#Target Variable statistics
motor_stats = {
   "mean": df["motor_UPDRS"].mean(),
   "median": df["motor_UPDRS"].median(),
   "std": df["motor_UPDRS"].std(),
   "min": df["motor_UPDRS"].min(),
   "max": df["motor_UPDRS"].max()
   }

total_stats = {
   "mean": df["total_UPDRS"].mean(),
   "median": df["total_UPDRS"].median(),
   "std": df["total_UPDRS"].std(),
   "min": df["total_UPDRS"].min(),
   "max": df["total_UPDRS"].max()
   }

#Demographics
Age_mean = df["age"].mean()
Range= {age_min == df["age"].min()} - {age_max == df["age"].max()}

#Gender Distribution
# Male =0 and Female =1
Male =  {(df['sex'] == 0).sum()}
Female = {(df['sex'] == 1).sum()}

#Recording Period
Duration = {df['test_time'].min()} {df['test_time'].max()}
Average_recordings_per_patient = {len(df) / df['subject#'].nunique()}
```
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The Summary Statistics for our Parkinsons Telemonitoring Dataset is as follows:
Target Variables (UPDRS Scores):
- Motor UPDRS: Mean = 21.30 (SD = 8.13), Range = [5.04, 39.51]
- Total UPDRS: Mean = 29.02 (SD = 10.70), Range = [7.00, 54.99]
- Both scores show substantial variability, indicating diverse disease severity across patients

Demographics:
- Age: Mean = 64.8 years, Range = 36-85 years
- Gender: 4,008 male recordings (68%) & 1,867 female recordings (32%)
- Recording Period: -4 to 215 days since recruitment

Voice Features:
- Jitter (%): Mean = 0.006 (measures pitch variation)
- Shimmer: Mean = 0.029 (measures amplitude variation)

- HNR: Mean = 21.68 dB (harmonic-to-noise ratio)
- All voice features show expected ranges for Parkinson's patients

C. Visualization

```
# ==============================================================================
# 4. VISUALIZATIONS
# ==============================================================================
#GENERATING VISUALIZATIONS

# Set style
sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = (16, 12)

# Create figure with 4 subplots
fig = plt.figure(figsize=(16, 12))

# Visualization 1: Distribution of Motor UPDRS Scores

ax1 = plt.subplot(2, 2, 1)
sns.histplot(df['motor_UPDRS'], bins=30, kde=True, color='steelblue', ax=ax1)
ax1.set_title('Distribution of Motor UPDRS Scores', fontsize=14,
fontweight='bold')
ax1.set_xlabel('Motor UPDRS Score', fontsize=12)
ax1.set_ylabel('Frequency', fontsize=12)
ax1.axvline(df['motor_UPDRS'].mean(), color='red', linestyle='--',
            linewidth=2, label=f'Mean: {df["motor_UPDRS"].mean():.2f}')
ax1.legend()

# Visualization 2: Correlation Heatmap

ax2 = plt.subplot(2, 2, 2)
voice_features = ['Jitter(%)', 'Shimmer', 'NHR', 'HNR', 'RPDE', 'DFA', 'PPE',
                  'motor_UPDRS', 'total_UPDRS']
correlation_matrix = df[voice_features].corr()
sns.heatmap(correlation_matrix, annot=True, fmt='.2f', cmap='coolwarm',
            center=0, square=True, linewidths=1, cbar_kws={"shrink": 0.8},
ax=ax2)
ax2.set_title('Correlation: Voice Features vs UPDRS', fontsize=14,
fontweight='bold')
plt.setp(ax2.get_xticklabels(), rotation=45, ha='right')

# Visualization 3: Disease Progression Over Time

ax3 = plt.subplot(2, 2, 3)
```

```
sample_patients = df['subject#'].unique()[:5]
for patient in sample_patients:
    patient_data = df[df['subject#'] == patient].sort_values('test_time')
    ax3.plot(patient_data['test_time'], patient_data['motor_UPDRS'],
             marker='o', markersize=3, alpha=0.7, label=f'Patient {patient}')
ax3.set_title('Motor UPDRS Progression Over Time (Sample)', fontsize=14,
fontweight='bold')
ax3.set_xlabel('Days Since Recruitment', fontsize=12)
ax3.set_ylabel('Motor UPDRS Score', fontsize=12)
ax3.legend(loc='best', fontsize=9)
ax3.grid(True, alpha=0.3)

# Visualization 4: Gender Comparison

ax4 = plt.subplot(2, 2, 4)
df['Gender'] = df['sex'].map({0: 'Male', 1: 'Female'})
sns.boxplot(x='Gender', y='motor_UPDRS', data=df, ax=ax4)
ax4.set_title('Motor UPDRS Distribution by Gender', fontsize=14,
fontweight='bold')
ax4.set_ylabel('Motor UPDRS Score', fontsize=12)
ax4.set_xlabel('Gender', fontsize=12)

plt.tight_layout()
plt.savefig('parkinsons_eda_visualizations.png', dpi=300, bbox_inches='tight')
plt.show()
```

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For the visualization I chose the following four graphs , which will help me with better insights of entire data and its trends. These graphs help me stick to my project's objective and give me better data interpretation. Further these graphs gave me a direction moving forward with my Project's Objective. The Four visualization and their outcomes are as follows:

1. Motor UPDRS Distribution (Histogram with KDE)
- Shows slightly right-skewed distribution
- Most patients cluster between 15-28 on motor UPDRS scale
- Mean line (red dashed) at 21.30 indicates central tendency
- Suggests varying levels of motor impairment across cohort

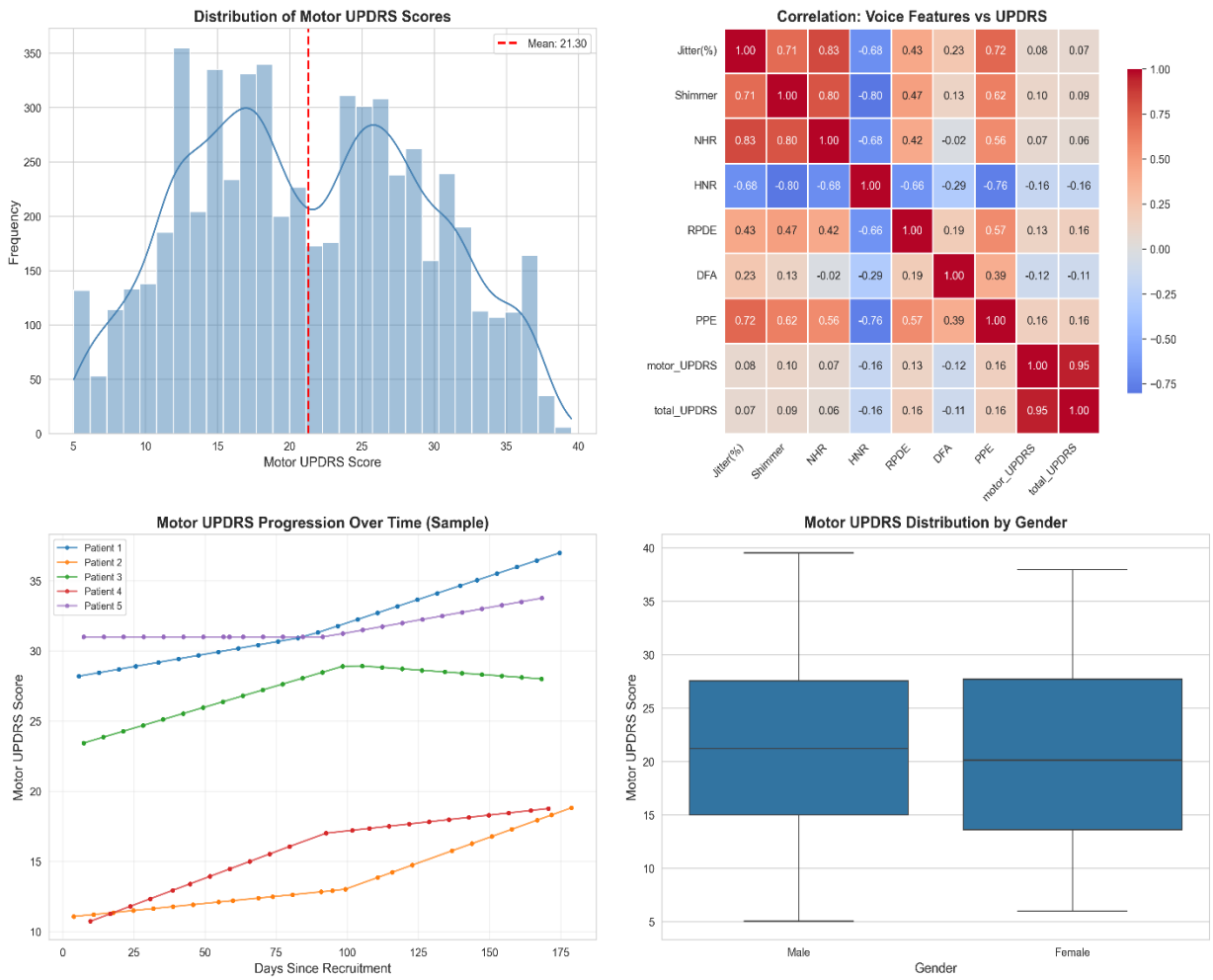
2. Correlation Heatmap (Voice Features vs UPDRS)
- Strong correlation between motor_UPDRS and total_UPDRS (0.95)
- PPE (nonlinear frequency variation) shows moderate positive correlation with UPDRS
- HNR (harmonic-to-noise ratio) shows slight negative correlation (-0.16)
- Multiple jitter and shimmer measures are highly intercorrelated (multicollinearity concern)

3. Motor UPDRS Progression Over Time (Line Plot)
- Sampled 5 patients to visualize disease progression
- Most patients show gradual increase in UPDRS scores over 6-month period

- Individual variability in progression rates
- Confirms time-series structure in data

4. Motor UPDRS by Gender (Box Plot)
- Male mean motor UPDRS: 21.47
- Female mean motor UPDRS: 20.92
- Minimal difference (0.54 points) - gender may not be a strong predictor
- Similar spread (IQR) between both groups

All visualizations have been saved as `parkinsons_eda_visualizations.png` (high-resolution 300 DPI).



D. <u>Key Findings:</u>

The Key Findings found from this analysis are as follows:

1. Data Quality is Excellent- The dataset is remarkably clean with no missing values, no duplicates, and consistent formatting. This allows me to focus on modeling rather than spending

excessive time on data cleaning. The UCI repository documentation matches the actual data structure perfectly.

2. Target Variable Shows Good Variability- Motor UPDRS scores range from 5 to 39.5, with a mean of 21.3 and standard deviation of 8.1. This variability is good for regression modeling—there is sufficient spread to learn meaningful patterns without extreme outliers dominating the predictions.

3. Voice Features Correlate Moderately with Disease Severity- The correlation heatmap reveals that voice features like PPE, Jitter, and Shimmer show moderate correlations with UPDRS scores. This supports the premise that voice recordings can track Parkinson's progression. However, correlations are not extremely strong (mostly 0.1-0.3 range), suggesting that regression models will need to capture complex, nonlinear patterns.

4. Multicollinearity is Present- Many voice features are highly correlated with each other (e.g., Jitter(%), Jitter:RAP, Jitter:DDP all measure pitch variation). This multicollinearity could cause issues for linear regression models. I will need to address this through:
- Regularization (Ridge/Lasso regression)
- Feature selection (drop redundant features)
- Dimensionality reduction (PCA)

5. Disease Progression is Evident- The time-series plot confirms that UPDRS scores generally increase over the 6-month recording period, consistent with Parkinson's being a progressive disease. This validates the dataset and suggests that time-series modeling approaches could be explored in future work (though I will keep it simple for this project).

6. Gender Differences are Minimal- Male and female patients show nearly identical motor UPDRS distributions (difference of 0.54 points). This suggests gender may not be a critical predictor in the final model, though I will keep it in the feature set and let the model decide.

E. <u>Challenges Encountered:</u>

1. Understanding Voice Features: The dataset includes 16 voice measurements with technical names like "Jitter:RAP" and "Shimmer: APQ5". I had to research what these features represent (pitch stability, amplitude variation, noise levels) to interpret the correlations meaningfully. The UCI documentation and reference papers helped clarify these.

2. Choosing Visualizations: With 22 features, deciding which visualizations would be most informative required thought. I prioritized:
- Target variable distribution (Motor UPDRS histogram)
- Feature-target relationships (correlation heatmap)
- Temporal patterns (progression line plot)
- Demographic comparisons (gender box plot)

3. Patient-Level Structure: The dataset has ~140 recordings per patient, creating a nested structure. For the train-test split in the next phase, I need to ensure entire patients go into either

training or testing—never split a single patient's recording across both sets. This is critical to avoid data leakage and overly optimistic performance estimates.

Conclusion and Next Plans:

As a fresher to machine learning, this EDA phase helped me understand the structure and scale of the Parkinsons Telemonitoring dataset before jumping into modeling. I learned that even though the dataset is relatively small in file size, it has a rich structure: repeated measures per patient, multiple correlated voice features, and clinically meaningful continuous targets (UPDRS scores).Visualizing the distribution of motor_UPDRS and its correlation with voice features gave me an intuitive sense that voice-based prediction is plausible but likely not perfectly linear, which will influence my choice of regression models in later weeks.

The main challenge I noticed is multicollinearity among the jitter and shimmer measures, so in the next phase I plan to explore feature scaling, regularization (ridge/lasso), and possibly dimensionality reduction to stabilize the models while keeping the interpretation as clear as possible.